

BTCS501

B.Tech. (CSE-BDA, CMC) IBM

VI Semester Examination, June-July, 2021

Theory of Computation

Choice Based Credit System (CBCS)

Time: 2 Hrs.

Maximum Marks: 60

Minimum Pass Marks: 24

- Note: 1. All questions carry equal marks, out of which part 'A' and 'B' carry 5 marks and part 'C' carries 6 marks.
 2. From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.
 3. Draw the neat diagram, wherever necessary.
 4. Assume suitable data, wherever necessary.
 5. Solve any three question

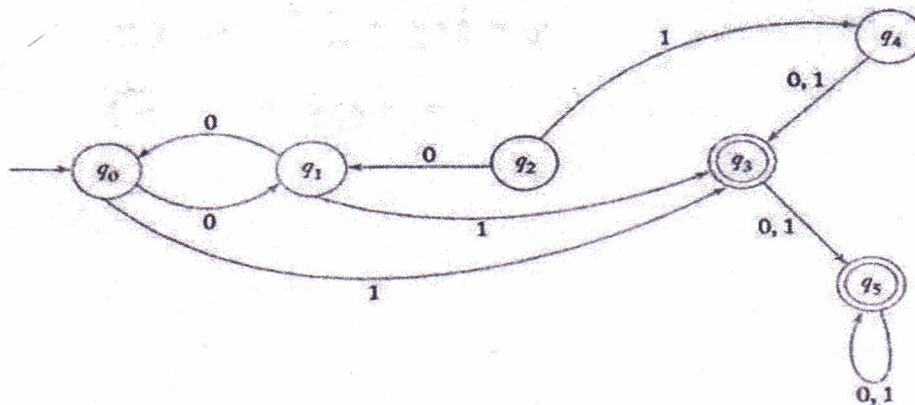
Q.1(A) Construct a finite automaton for the language $L = \{0^n | n \bmod 3 = 2, n \geq 0\}$

05

(B) Construct a DFA for the language $L = \{a^n : n \text{ is a multiple of 3 or a multiple of 5}\}$

05

(C) Minimize the states in the DFA depicted in the following diagram.

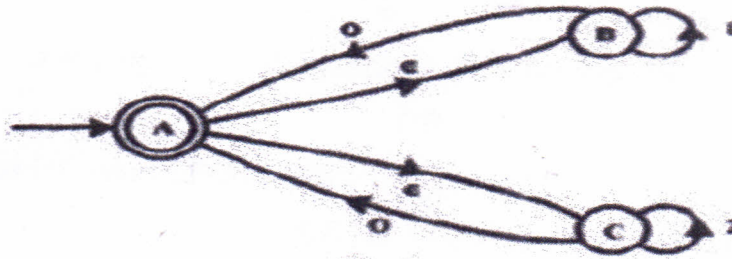


OR

10

Contd...

Convert the following NFA with ϵ moves to equivalent DFA.



Q.2(A) Prove that the following language is not regular by using pumping lemma.

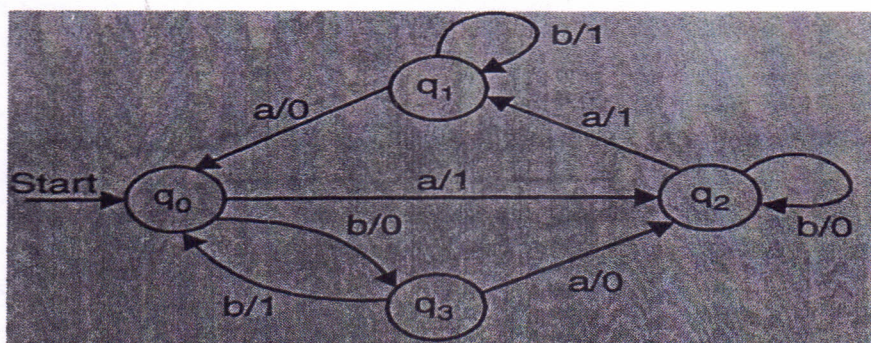
$\{0^n : n \text{ is perfect square}\}$

05

(B) Find the regular expression for the language of all string containing at most two 0's.

05

(C) Convert the following mealy machine into moore machine.



OR

Find regular expression for the following language:

$L = \{w : n_a(w) \text{ and } n_b(w) \text{ are both even}\}$

10

Q.3(A) Show that the grammar

$S \rightarrow aB \mid ab$

$A \rightarrow aAB \mid a$

$B \rightarrow ABb \mid b$

Is ambiguous.

05

(B) Eliminate all λ -productions from:

$S \rightarrow AaB \mid aaB$

$A \rightarrow \lambda$

$B \rightarrow bbA \mid \lambda$

05

Contd...

- (C) The following grammar generates the language of regular expression $0^*1(0+1)^*$:

$S \rightarrow AB$

$A \rightarrow 0A \mid \epsilon$

$B \rightarrow 0B \mid 1B \mid \epsilon$

Give leftmost and rightmost derivation of the following string: 00101

OR

Convert the following CFG to GNF:

$A \rightarrow BC$

$B \rightarrow CA \mid b$

$C \rightarrow AB \mid a$

10

Q.4(A) Give the definition of Pushdown Automaton with the help of diagram.

05

(B) Define Deterministic PDA.

05

(C) Design a PDA for the language: $L = \{w \in (a+b)^* \text{ and } n_a(w) > n_b(w)\}$

OR

Design a Pushdown Automaton for balanced parentheses. $\{\{\{\}\}\}\}$

10

Q.5(A) Explain multi tape Universal Turing Machine.

05

(B) Explain the Off-Line Turing Machine.

05

(C) Construct a Turing machine that will accept the following language on $\{a,b\}$. $L = \{w : |w| \text{ is even}\}$.

OR

Prove that the set of all languages that are not recursively enumerable is not countable.

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